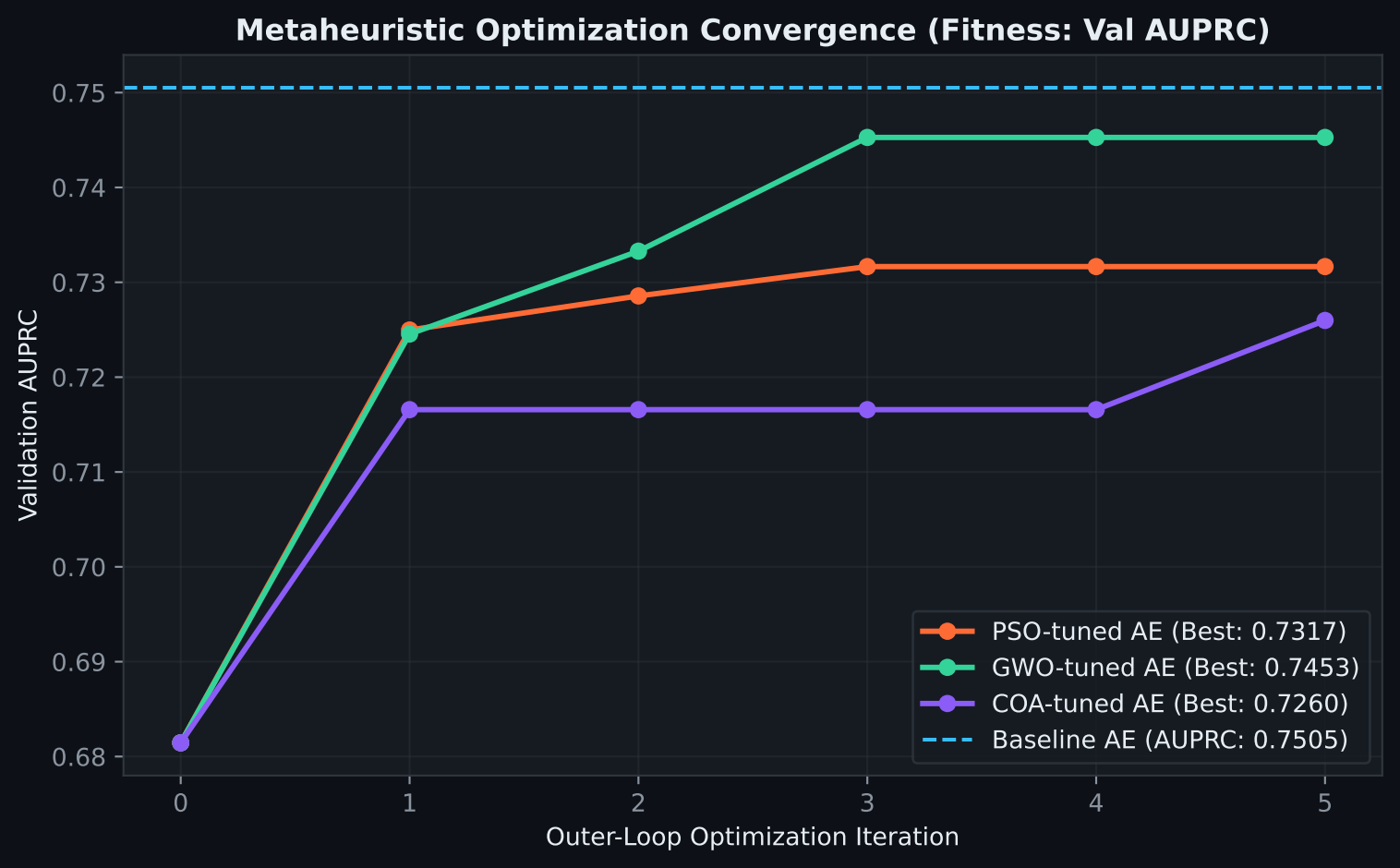
Metaheuristic Optimization Convergence (Fitness: Val AUPRC)
Validation AUPRC
0.75
0.74
0.73
0.72
0.71
0.70
0.69
0.68
0
1
2
3
4
5
Outer-Loop Optimization Iteration
PSO-tuned AE (Best: 0.7317)
GWO-tuned AE (Best: 0.7453)
COA-tuned AE (Best: 0.7260)
Baseline AE (AUPRC: 0.7505)